The Hexagonality Law in Real Materials III

Alloys and hydrogen

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. Hydrogen is the enemy of titanium implants and of the zirconium tubes that hold nuclear fuel: it gathers in the metal and turns it brittle. We show that the places hydrogen can sit follow the Hexagonality Law (#66) exactly. In a close-packed metal of any stacking, per metal atom, the pairs of tetrahedral holes that share a face, and the pairs of octahedral holes that share a face, both number h, the share of hexagonal layers. Two hydrogen atoms cannot sit in one face-sharing tetrahedral pair, so at most $2 - h$ tetrahedral sites per atom can be filled, and a hydride MH_x with hydrogen in tetrahedral sites needs the metal locally to have $h \leq 2 - x$: the brittle dihydrides need fully cubic stacking. DICTO proposed to test the law on alloys, starting from Ti-6Al-4V and Ti-Nb-Zr. Across six alloy families the law sorts them on one scale: exact for the hole and hydrogen counts; the composition of the magnesium LPSO alloys is a straight line in h in the authors' own model; the energies of cobalt and iron stackings come close (second-neighbour share 0.15–0.17); the faults of the high-entropy alloy CrCoNi depart clearly. In the standard fault model of these steels and alloys, Olson and Cohen's interface energy σ is exactly twice the second-neighbour layer coupling, so it measures the distance from the law.

1. Why alloys

Papers #67–#70 tested the Hexagonality Law on stackings of single elements and simple compounds. Engineering metals are alloys, and their most important stacking events are faults: the turning of a few layers from hexagonal to cubic or back. DICTO proposed to test the law there, starting from the titanium alloys Ti-6Al-4V (aircraft, implants, printed parts) and Ti-Nb-Zr (low-stiffness implants), and to include their spherical powders. We added four families in which cubic against hexagonal stacking is the main story: the long-period stacking (LPSO) magnesium alloys, Fe–Mn and high-entropy alloys, cobalt alloys, and the zirconium alloys of nuclear fuel cladding. A powder bed or a printed part is made of grains; inside each close-packed grain the results below apply unchanged.

2. Hydrogen sites

Take balls of diameter a in close-packed layers A, B, C, any stacking, and let h be the share of hexagonal layers (a layer is hexagonal when the layers above and below it sit in the same position). Every stacking has 2 tetrahedral and 1 octahedral hole per ball (#66).

Theorem. Per ball, in any close-packed stacking: (i) the pairs of tetrahedral holes sharing a face number h; (ii) the pairs of octahedral holes sharing a face number h. Consequently (iii) if two hydrogen atoms cannot occupy one face-sharing tetrahedral pair, at most $2 - h$

> tetrahedral sites per ball can be filled, and (iv) a hydride MH_x with its hydrogen in tetrahedral sites needs local hexagonality $h \le 2 - x$.

Proof. (i) A tetrahedral hole has its base on a triangle of one layer and its apex on a ball of the next. Two tetrahedra can share a face only across a layer i, on the same triangle, one with its apex in layer $i - 1$ and one in layer $i + 1$. For each ball of layer i there is exactly one triangle of i that lies under a ball of $i + 1$, and its partner below exists exactly when layer $i - 1$ sits where $i + 1$ sits, that is when layer i is hexagonal. So each hexagonal layer gives one pair per ball and each cubic layer none: h pairs per ball. (ii) The octahedral holes between two layers sit over the third position, the one neither layer uses. The holes between $i - 1$ and i and between i and $i + 1$ share a face of layer i exactly when they sit over the same position, that is when $i - 1$ and $i + 1$ coincide: again one pair per ball of each hexagonal layer. (iii) Each pair loses one of its two sites, and the pairs are disjoint (each tetrahedron has one base layer), so $2 - h$ remain. (iv) follows. ∎

We checked (i) and (ii) on explicit coordinates for six stackings (3C, 2H, 4H, 6H and the 10H and 14H of §3), periodic in all three directions; all counts are exact. The same run finds the tetrahedral pairs that share only an edge (centres 0.707a apart) at $6 - 3h$ per ball, also a straight line (computed, not proved here).

A face-sharing tetrahedral pair has its centres 0.408a apart: 1.20 Å in α-titanium (a = 2.95 Å), 1.32 Å in α-zirconium (a = 3.23 Å), far inside the 2.1 Å that hydrogen atoms in metals keep apart (Switendick's criterion), so (iii) applies to both. The edge-sharing pairs are 2.28 Å apart in zirconium (allowed) and 2.09 Å in titanium (at the limit; no claim made). Real titanium and zirconium are slightly flattened (c/a ≈ 1.59 against the ideal 1.633): the counts are exact, the distances shift by about 3%.

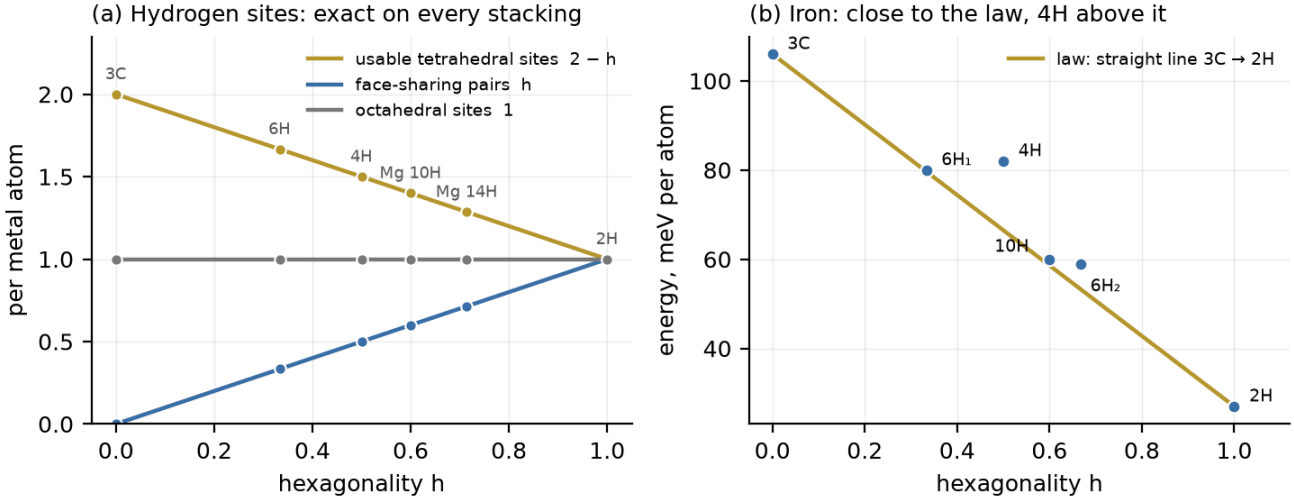


Figure 1. (a) Hydrogen sites per metal atom against the hexagonality, from explicit coordinates: usable tetrahedral sites $2 - h$, face-sharing pairs h, octahedral sites 1, exact on every stacking tested. (b) Energies of six iron stackings (non-magnetic, Tsumuraya et al.) against the law's straight line through 3C and 2H: close, with 4H above the line.

3. Magnesium LPSO alloys

The Mg–Zn–Y alloys owe their strength to long-period stacking ordered phases: hexagonal magnesium with a fault every k layers, each fault making two cubic layers that collect the zinc and yttrium (Abe, Ono, Itoi, Yamasaki and Kawamura, from atom-resolved microscopy). Built from their description and checked with Jagodzinski's rule:

Phase	fault every	stacking (one period)	h
10H	5 layers	ABCBCBACAC	3/5
18R	6 layers	ABCBCBCABABABCACAC	2/3
14H	7 layers	ABCBCBCBACACAC	5/7
24R	8 layers	ABCBCBCBCABABABABCACACAC	3/4

Abe and co-workers estimate the compositions by giving each cubic layer the same composition $Mg_{85}Zn_5Y_{10}$ and each hexagonal layer pure magnesium, and their microscopy shows the same cubic-layer intensity in every phase. That model is the law's form: the zinc-plus-yttrium content is 15% × (1 − h), giving their 6.0, 5.0, 4.3 and 3.7%. By the Theorem, the hydrogen-site counts of these phases are straight lines in the same h.

4. Faults and the second neighbour

In the layer (ANNNI) model used in #68, a property relative to cubic stacking is $X(2H) = 2(J_1 + J_3)$ per layer, and the fcc faults cost $ISF = 4(J_1 + J_2 + J_3)$, $ESF = 4(J_1 + 2J_2 + 2J_3)$, twin $2(J_1 + 2J_2 + 3J_3)$ (Denteneer and van Haeringen). The law holds exactly when $J_2 = J_3 = 0$ (#68). Steels and alloys that deform by turning cubic layers hexagonal (TRIP) or by twinning (TWIP) are described by Olson and Cohen's formula, $ISF = 2\rho\Delta G + 2\sigma$, where $\rho\Delta G$ is the cost per area of one hexagonal layer and σ an interface energy.

> **Proposition.** In the layer model, $\rho\Delta G = 2(J_1 + J_3)$ and $2\sigma = 4J_2$. So $\sigma = 0$ exactly when the second-neighbour coupling vanishes, and σ measures how far a material's faults are from the law.

Proof. $\rho\Delta G$ is $X(2H)$ per layer, $2(J_1 + J_3)$; subtract twice that from $ISF = 4(J_1 + J_2 + J_3)$. ∎ (An immediate consequence of the standard expressions; not claimed as new.)

Material (data)	what was tested	result	$\lvert J_2/J_1 \rvert$
hcp cobalt (Aas et al., four computed faults)	energy per cubic layer made: I1 16, T2 39, I2 48, E 62 meV for 1, 2, 2, 3 layers	16.0, 19.5, 24.0, 20.7 meV: within ±20% of one constant	0.15
iron, six stackings (Tsumuraya et al., non-magnetic)	energy and volume against h	volume linear within 0.1 $Bohr^3$ of 69 (rounding); energy close, 4H 12 meV/atom above the line	0.17
CrCoNi high-entropy alloy (Ding et al.)	ISF against ESF (the law says equal)	random −42.9 / −27.8, ordered 30 / 66 mJ/m^2; $2\sigma \approx 18$ mJ/m^2	≈ J_1-sized

Cobalt and iron are close to the law: about four fifths of a fault's cost is the per-cubic-layer term. The 10H of iron in that set is the same stacking as the magnesium 10H of §3. In CrCoNi the local chemical order reaches across layers and the law's prediction ESF = ISF misses by 15 to 36 mJ/m^2.

5. Hydrides of zirconium and titanium

By the Theorem (iv):

Hydride	x	needs h ≤
γ-ZrH	1	1 (the count allows hexagonal)
δ-$ZrH_{1.5}$	1.5	0.50
δ-$ZrH_{1.66}$	1.66	0.34
ε-ZrH_2, δ-TiH_2	2	0 (fully cubic)

The hydrogen-rich hydrides that embrittle zirconium cladding and titanium therefore need the metal locally at least two-thirds cubic. This agrees with the accepted mechanism, in which the metal turns from hexagonal to cubic by partial dislocations gliding on alternate basal planes (Carpenter), each leaving a fault that makes cubic layers (#67). The bound is a necessary condition only: γ-ZrH is itself cubic-stacked though the count would allow hexagonal.

6. Where the law stops

Body-centred cubic metals are not close-packed stackings, and the law does not apply to them: the β phase of Ti-Nb-Zr, ferritic steels such as chromium–molybdenum steel, and the ω phase of titanium alloys. The α″ martensite of Ti-Nb-Zr is a distorted hexagonal stacking: the counts of §2 apply to it, the distances shift. The cobalt–chromium–molybdenum alloys of implants are close-packed and belong with cobalt in §4.

7. Conclusion

The law sorts alloys on one scale. Exact: the hole and hydrogen counts of every stacking, hence the bound $h \le 2 - x$ for hydrides. Exact in the authors' own model: the composition of the magnesium LPSO phases. Close: the stacking energies of cobalt and iron. Broken: the faults of the high-entropy alloy CrCoNi, where Olson and Cohen's σ, twice the second-neighbour coupling, is large.

8. Prior art and credit

The LPSO stackings and the composition model are Abe, Ono, Itoi, Yamasaki and Kawamura's; the iron polytype energies and volumes are Tsumuraya, Watanabe and Sawaguchi's; the cobalt fault energies are Aas, Szunyogh, Evans and Chantrell's; the CrCoNi fault energies are Ding, Yu, Asta and Ritchie's; the fault model is Olson and Cohen's; the layer-model fault expressions are Denteneer and van Haeringen's; the hydride mechanism is Carpenter's; the hydrogen distance criterion is Switendick's. That face-sharing tetrahedral pairs occur around hexagonal layers is standard crystallography; the count h per ball for any stacking, the resulting bound for hydrides, and the reading of σ as $2J_2$ are stated here with proofs, and the proofs are short. DICTO proposed the test on alloys and the titanium cases; the further families and the computations are joint work.

> **Sources checked** (28 October 2026). Read: E. Abe, A. Ono, T. Itoi, M. Yamasaki and Y. Kawamura, Polytypes of long-period stacking structures synchronized with chemical order in a dilute Mg-Zn-Y alloy, arXiv 1106.4944 (pp. 3–5); T. Tsumuraya, I. Watanabe and T. Sawaguchi, Origin of phase stability in Fe with long-period stacking order…, arXiv 2010.07118 v4 (Tables I–II; in the reference library); C. J. Aas, L. Szunyogh, R. F. L. Evans and R. W. Chantrell, Effect of stacking faults on the magnetocrystalline anisotropy of hcp Co, arXiv 1304.1353 (pp. 3, 7); J. Ding, Q. Yu, M. Asta and R. O. Ritchie, Tunable stacking fault energies by tailoring local chemical order in CrCoNi, arXiv 1806.00718 (pp. 9–13). Cited as given in those papers, not read: G. B. Olson and M. Cohen, Metall. Trans. A 7, 1897 (1976); P. J. H.

Denteneer and W. van Haeringen, J. Phys. C 20, L883 (1987); G. J. C. Carpenter, J. Nucl. Mater. 48, 264 (1973); A. C. Switendick, Z. Phys. Chem. N.F. 117, 89 (1979). Search summaries, not read in full: arXiv 1811.12442 (δ-hydrides in Zircaloy-4); PMC10532661 (fatigue-induced fcc zirconium); Ikeda et al., Entropy 2018 (PMC7513178); Lizárraga et al., Sci. Rep. 2017 (cobalt hcp–fcc). Web searches: Mg-Zn-Y LPSO stacking sequences; stacking fault energies of CoCrNi, CoCrFeMnNi and Fe-Mn; cobalt fault energies; δ-hydride hcp to fcc mechanism. Library searched for each name above.

Data and code. Study 2026-10-28-law-in-alloys (lpso.py, hydrogen_sites.py, hea_faults.py, cobalt_faults.py, iron_polytypes.py, hydride_bound.py, fig71.py, with outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275; The Hexagonality Law in Real Materials II (#70), 10.5281/zenodo.23723565.